SCC.160: Fundamentals of Communication Systems
Lecture 4

Hasan Ahmed
h.ahmed@lancaster.ac.uk

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Overview

- Bit Rate vs. Baud Rate / Symbol Rate
- Data Rate Limits
- Multi-level Digital Modulation
  - M-ASK
  - M-FSK
  - M-PSK
  - QAM

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Bit Rate vs. Baud Rate / Symbol Rate

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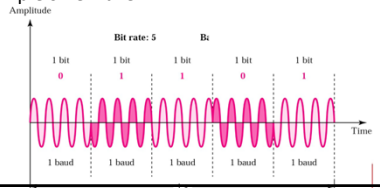
Digital Modulation

- In a digital modulation method provided by a modem, **each symbol is typically a sine wave tone with a certain frequency, amplitude and phase**. Symbol rate, baud rate, is the number of transmitted tones per second.
- One symbol can carry one or several bits of information.
- System is time-slotted and only send ‘0’ and ‘1’ within a time-slot
- In each time slot we transmit a simple sine wave

Bit	Phase
0	0
1	180

Bits

Constellation diagram



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N is bit rate
S is baud rate
r is the number of bit per symbol.

Baud rate = Symbol rate

Bit Rate vs. Baud Rate / Symbol Rate

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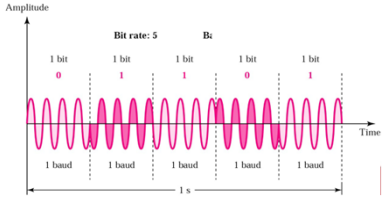
- We have previously seen that digital data is specified by a bit rate
- When the data bits are translated to analogue (sine waves), "they become symbols"
- The bits are the original digital data stream. The symbols (pink dots) are what we actually transmit!

Bit	Phase
0	0
1	180

Bits

Constellation diagram



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Bit Rate vs. Baud Rate / Symbol Rate

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- Bit rate: the number of bits sent in one second, unit: bps
- Baud rate: the number of signal elements per second
  - Each signal element is commonly referred to as a symbol.
  - In digital modulation, **each symbol is typically a sine wave with a certain frequency, amplitude and phase** that persists for a fixed period of time.
- Baud rate is also known as symbol rate or modulation rate.
  - 'Symbol rate' is **more commonly used** in modern wireless communication area.
  - Unit of symbol rate: baud (bd) or **symbols per second**.
  - We will use these terms interchangeably.
- Relationship between symbol rate and bit rate: N = r * S
  - N is the bit rate, S is the baud rate or symbol rate, r is the number of bits per symbol and $r=\log_2(L)$, where L is the number of different signal elements.

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Bit Rate vs. Baud Rate / Symbol Rate

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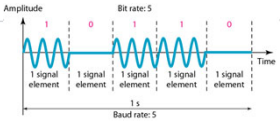
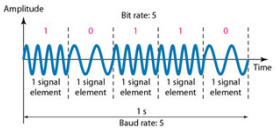
对于二进制调制: bit rate = symbol rate

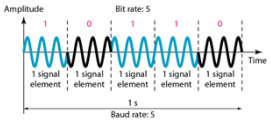
BASK



BFSK



BPSK



For **Binary modulation schemes**, i.e., BASK, BFSK, BPSK, symbol rate = bit rate.

- N = r * S where $r=\log_2(L)$ and L is the number of different signal elements (symbols)
- L = 2 for binary modulation methods
- Hence, r = 1 and N=S. It means **BIT RATE = SYMBOL RATE**

$r=\log_2 L$
L为电平数

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Data Rate Limits

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$B = (1+d)S$

$N = 2B\log_2 L$ → Nyquist law
$N = S\log_2 L$ → bit & baud
∴ $S = 2B$, S为理想最大波特率 = 2B (bandwidth)

Maximum Data Rate

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Data Rate Limits: how fast we can send data in bits per second over a channel?

Noiseless channel

- The theoretical maximum bit rate for a certain spectral bandwidth B (in hertz) is given by *Nyquist law*: $2B\log_2 L$, where B is the channel bandwidth, L is the number of different symbols.
- Since $N = S\log_2 L$, it also means that the maximum achievable symbol rate is 2B, **twice the bandwidth**.
  - Can we use arbitrarily large value of L? No. When L increases, it becomes more difficult to distinguish the symbols at the receiver.

Noisy channel

- The theoretical maximum data rate for a noisy channel subject to AWGN noise is given by Shannon capacity: $C = B\log_2(1 + S/N)$, where C is the channel capacity in bps, B is the channel bandwidth in hertz, S is the average received signal power in watts, N is the average noise power (including interference if any). S/N is the SNR, expressed in linear power ratio, not in dB.

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Quiz

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- Q: Consider an extremely noisy channel in which the value of the signal-to-noise ratio is almost zero. In other words, the noise is so strong that the signal is faint. What is the channel capacity for this channel?
- Solution:
- For this channel, the channel capacity C is calculated as

$$C = B\log_2(1 + \text{SNR}) = B\log_2(1 + 0) = B\log_2 1 = B \times 0 = 0$$

This means that the capacity of this channel is zero regardless of the bandwidth. In other words, we cannot receive any data through this channel.

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Quiz

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- Q: We can calculate the theoretical maximum bit rate of a regular telephone line. A telephone line normally has a bandwidth of 3000 Hz. The signal-to-noise ratio is usually 3162. What is the channel capacity?
  B
- Solution:
- For this channel, the channel capacity is calculated as

$$C = B\log_2(1 + \text{SNR}) = 3000\log_2(1 + 3162) = 3000\log_2 3163$$
$$= 3000 \times 11.62 = 34860 \text{ bps}$$

This means that the theoretical maximum bit rate for a telephone line is 34.860 kbps. If we want to send data faster than this, we can either increase the bandwidth of the line or improve the signal-to-noise ratio.

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Multi-Level Digital Modulation

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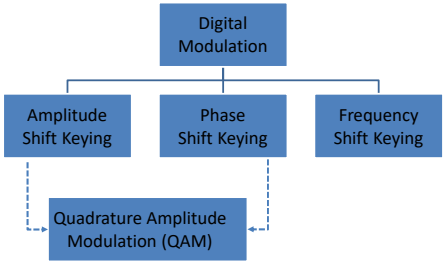
Digital Modulation

- A sine wave is defined by three characteristics: amplitude, frequency, and phase.
- Any of the three characteristics can be altered in this way, giving us at least three mechanisms for modulating digital data into an analogue signal: amplitude shift keying (ASK), frequency shift keying (FSK), and phase shift keying (PSK)



*In addition, there is a fourth (and better) mechanism that combines changing **both the amplitude and phase**, called quadrature amplitude modulation (QAM). QAM is the most efficient of these options and is the mechanism commonly used today.*

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$L = 2^4$

4 bits

$\underbrace{_\ _\ _\ _}_{\text{symbol}}$

$r = 4$

Multi-level Digital Modulation 多级数字调制

- Binary digital modulation (BASK, BFSK, BPSK): transmitting one bit at a time by employing two signal elements, i.e., L = 2. 0 和 1
- Multi-level digital modulation (or called ***M-ary modulation***): simultaneous transmission of multiple bits by using more than two signal elements
  - Each symbol represents **more than one bit**
  - There are in total $L = 2^r$ signal elements (symbols), each of duration T_s
  - Symbol rate S = 1/ T_s
  - Each symbol carries r=$\log_2$(L) bits of information
  - Bit rate N = r*S = r/ T_s
  - M-ary modulation allows to transmit a higher bit rate with a given bandwidth, compared to binary digital modulation
    - Because it can keep the symbol rate the same and increase the number of bits per symbol

一共有2^r个电平，每个都有Ts的持续时间

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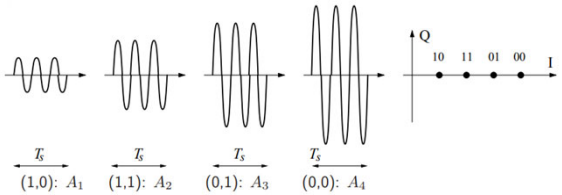
对于BASK，由于一次只能传输0和1，L=2，因此效率低下。

Multi-level ASK (M-ASK)

- There are **multiple different amplitude values of the carrier**.
- Each amplitude encodes an equal number of bits (more than one bit).
- Each pattern of bits forms the symbol that is represented by a signal element with a specific amplitude.
- An example of 4ASK signal elements and a constellation diagram



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Multi-level ASK (M-ASK)

- The transmitted MASK signal can be represented by ($i = 1,2,\ldots,L$)

$$s_i(t) = A_i \cos(2\pi f_c t + \phi),\ 0 \le t \le T_s$$

- Recall that in ASK, the frequency and phase remain constant, while only amplitudes are used to represent information. Hence, only A_i varies.
- Example, let L = 4. The four signal amplitudes could be -3, -1, 1 and 3 V. The 4-ASK signals will be

$$s_1(t) = \cos(2\pi f_c t + \phi),\ 0 \le t \le T_s$$
$$s_2(t) = -\cos(2\pi f_c t + \phi),\ 0 \le t \le T_s$$
$$s_3(t) = 3\cos(2\pi f_c t + \phi),\ 0 \le t \le T_s$$
$$s_4(t) = -3\cos(2\pi f_c t + \phi),\ 0 \le t \le T_s$$

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正交相移键控：QPSK
L=4, r=2,

Multi-level FSK (M-FSK)

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- In the case of M-FSK, there are **L different carrier frequencies** used to represent multiple symbols.
- The frequencies must be chosen carefully such that there is no interference between adjacent frequency carriers.
- Quiz: Suppose we need to send 3 bits at a time and the bit rate is 3 Mbps. Consider FSK. Calculate the number of different frequencies or symbols and symbol rate.
- Solution: It indicates that r = 3, $L = 2^r$ = 8. Then, there are 8 different frequencies required. The symbol rate = bit rate / the number of bits per symbol = 3 Mbps / 3 = 1 Mbaud.

S N r

$L = 2^3 = 8$

$N = S \cdot r$

$3\times10^6 = S \cdot 3 = 1M$ baud

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Multi-level PSK (M-PSK)

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- In the case of MPSK, there are **L different carrier phase shifts** used.
- Each symbol is represented by one of the phase shifts.
- Example: instead of transmitting just 2 possible phase shifts (0˚and 180˚), we could transmit one of 4 possible phase shifts per symbol. This is called *quadrature phase shift keying (QPSK or called 4PSK).* 正交相移键控 L=4
- In QPSK, there are four symbols (L=4) and two bits per symbol (r= $log_2 L$ =2) r=2
- QPSK can either **double the data rate** compared with a BPSK system while maintaining the same bandwidth of the signal, or to maintain the data-rate of BPSK but halving the bandwidth needed.

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Multi-level PSK (M-PSK)

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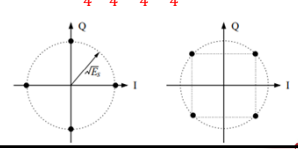
- M-PSK signal set ($i = 1,2,3,\ldots,L$):

$$s_i(t) = A_i \cos(2\pi f_c t + \phi),\ 0 \le t \le T_s$$

- where the phase ϕ_i is typically given by $\phi_i = \frac{2\pi}{L}(i-1)$ + constant
- For QPSK (L=4), if the constant is 0, then the four phases are 0, $\frac{\pi}{2}$, π, $\frac{3\pi}{2}$.
- For QPSK (L=4), if the constant is $\frac{\pi}{4}$, the four phases are $\frac{\pi}{4}$, $\frac{3\pi}{4}$, $\frac{5\pi}{4}$, $\frac{7\pi}{4}$.
- Both are equivalent variations of QPSK.
- QPSK constellation diagram
  - Minimum phase separation is 90°

最小的分离相度为90°



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Multi-level PSK (M-PSK)

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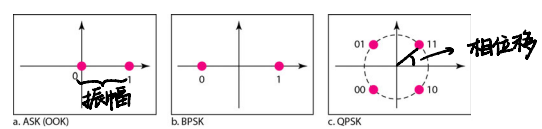
- The transmitted QPSK signal can also be written as
- $Acos(2\pi f_c t + \phi_i) = Acos(\phi_i)\cos(2\pi f_c t) + (-Asin(\phi_i))\sin(2\pi f_c t)$

In-phase component Quadrature-phase component

- Note in QPSK, both in-phase and quadrature branches (two orthogonal carriers) are utilised



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Multi-level PSK (M-PSK)

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- The examples below show the signal constellations of BPSK and 4PSK respectively
- In BPSK, the 0 bit is mapped to a symbol of zero phase and the 1 bit is mapped to a symbol of 180° phase shift
- In QPSK, each symbol represents two bits. Each symbol is represented by one of the four phase shifts: 0°, 90° , 180° , 270°.

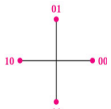
Bit	Phase
0	0
1	180

Bits

Constellation diagram

Dibit	Phase
00	0
01	90
10	180
11	270

Dibit
(2 bits)



Constellation diagram

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Multi-level PSK (M-PSK)

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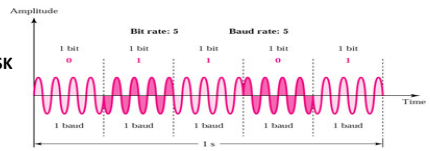
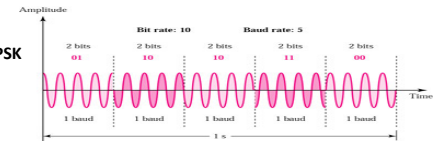
N=5 N=S·r
r=1 S=5

对于BPSK, N=S

BPSK/2PSK



QPSK/4PSK



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Multi-level PSK (M-PSK)

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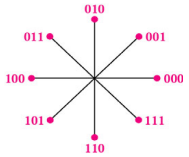
- Higher-order PSK: further increase the number of bits per symbol by increasing the number of phase shifts.
- 8-PSK: 8 different carrier phase shifts
- Quiz: how many bits per symbol are transmitted in 8-PSK?
- Solution: since there are 8 symbols, i.e., L=8, so r = $log_2 L$ = 3 bits per symbol.

Tribit	Phase
000	0
001	45
010	90
011	135
100	180
101	225
110	270
111	315

Tribits
(3 bits)



Constellation diagram

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Multi-level PSK (M-PSK)

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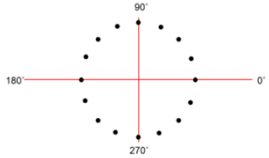
- 16-PSK: further increase to 4 bits/symbol by using 16 different carrier phase shifts, i.e., L = 16 and r = 4 bits/symbol.
- A possible 16-PSK constellation is shown below, where each phase is separated by 360° /16 = 22.5^o.
  - Recall: for PSK, all the symbols have the same carrier frequency and amplitude; it is their phase that is different. Therefore, on a PSK constellation diagram, all the symbols appear on a circle about the origin.



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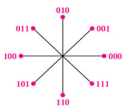
Multi-level PSK (M-PSK)

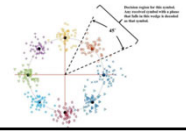
- Can we use arbitrarily complex PSK modulations, e.g., 64-PSK, 128-PSK?
  - It becomes more susceptible to noise as the symbols get closer together.
  - As we introduced before, the received signal will be degraded by noise.
  - The noise corruption can make it harder for a receiver to determine the correct phase (correct symbol) that was transmitted, especially when more phases are used in PSK making the symbols closer to each other.
  - For example, in 8PSK, when the noise is severe enough, a receiver may make a mistake and decide that it had received 011 but actually it should be 010. These are called bit errors.
  - There is a way to use more symbols in modulation while reducing the chances of making bit errors; by using symbols that have different amplitudes AND phases so they are more spread out.

Tribit	Phase
000	0
001	45
010	90
011	135
100	180
101	225
110	270
111	315

Tribits (3 bits)



Constellation diagram



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Quadrature Amplitude Modulation (QAM)

QAM: 同时对振幅和相位进行调制

- In order to increase the distance between symbols in a constellation, we can modulate both the *amplitude* and the *phase* of the carrier. This is called Quadrature Amplitude Modulation (QAM). 正交调幅
- QAM is a popular bandwidth-efficient modulation scheme used in many practical systems.
- In QAM, both a sine and a cosine are simultaneously used for transmitting information. The cosine part is called the in-phase component and the sine part is called the quadrature-phase component.

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Quadrature Amplitude Modulation (QAM)

- If different (multiple positive and negative) amplitudes are used with the two phaseshifted carriers (sine and cosine), the modulation scheme is called quadrature amplitude modulation (QAM).
- The general M-QAM signal can be written as ($i = 1,2,3,\ldots,L$):

$$s_i(t) = A_{i,I}\cos(2\pi f_c t) + A_{i,Q}\sin(2\pi f_c t),\ 0 \le t \le T_s$$

- where the subscripts I and Q refer to the in-phase and quadrature-phase components. Note that we can also write the QAM signal as ($i = 1,2,3,\ldots,L$):

$$s_i(t) = A_i\cos(2\pi f_c t + \phi_i),\ 0 \le t \le T_s$$

where $A_i = \sqrt{A_{i,I}^2 + A_{i,Q}^2}$ and $\phi_i = -\tan^{-1}\left(\frac{A_{i,Q}}{A_{i,I}}\right)$.

- So we can think of QAM as a mix of both amplitude and phase shift keying since *the i-th symbol is mapped to a carrier with amplitude A_i and phase ϕ_i*.

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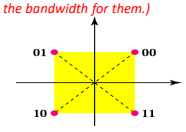
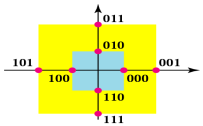
Quadrature Amplitude Modulation (QAM)

- 4QAM: identical to QPSK (Although the root concepts of QPSK and 4-QAM are different, the resulting modulated radio waves are exactly the same.)
- 8QAM: an 8QAM constellation diagram is shown below (one of many possible 8QAM constellations).
  - 8 symbols, each representing 3 bits
  - 2 amplitudes and 4 phases are used
- What is the bandwidth for 8QAM? The same as 8PSK. *(*You are not expected to know how to calculate the bandwidth for them.)*



4-QAM
1 amplitude, 4 phases



8-QAM
2 amplitudes, 4 phases

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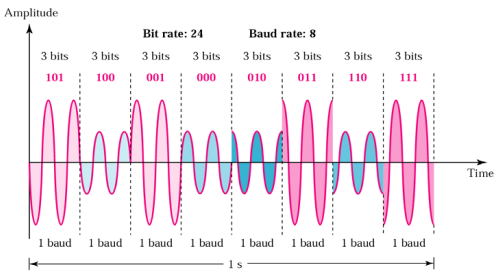
Quadrature Amplitude Modulation (QAM)



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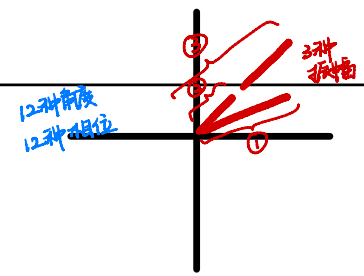
Quadrature Amplitude Modulation (QAM)

- Higher-order QAM
  - QAM can be extended to have a larger number of symbols, which then allows a much higher bit rate in the same bandwidth (because there are more bits per symbol).
  - In QAM, the constellation points are usually arranged in a square grid with equal vertical and horizontal spacing.
  - The most common QAM forms are 16-QAM, 64-QAM, 256-QAM.
  - 64-QAM and 256-QAM are often used in digital cable television and cable modem applications.



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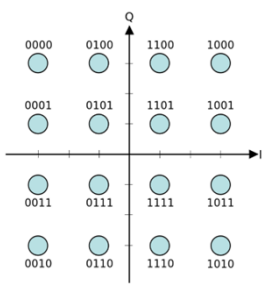
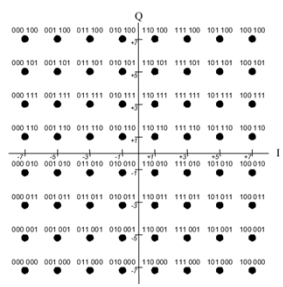
Quadrature Amplitude Modulation (QAM)

- Some possible constellations are given below.



16-QAM



64-QAM

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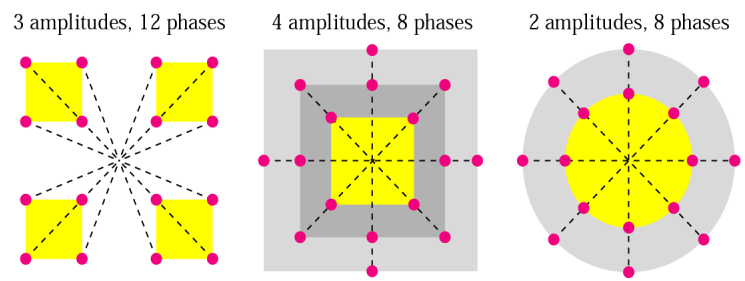
Some Possible Constellations



16-QAM 16-QAM 16-QAM

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Bit and Symbol Rate Comparison

Modulation	Bits/symbol	Symbol rate	Bit Rate
BASK, BFSK, BPSK	1	x	x
4-PSK, 4-QAM	2	x	2x
8-PSK, 8-QAM	3	x	3x
16-PSK, 16-QAM	4	x	4x
32-QAM	5	x	5x
64-QAM	6	x	6x
128-QAM	7	x	7x
256-QAM	8	x	8x

Thank You!